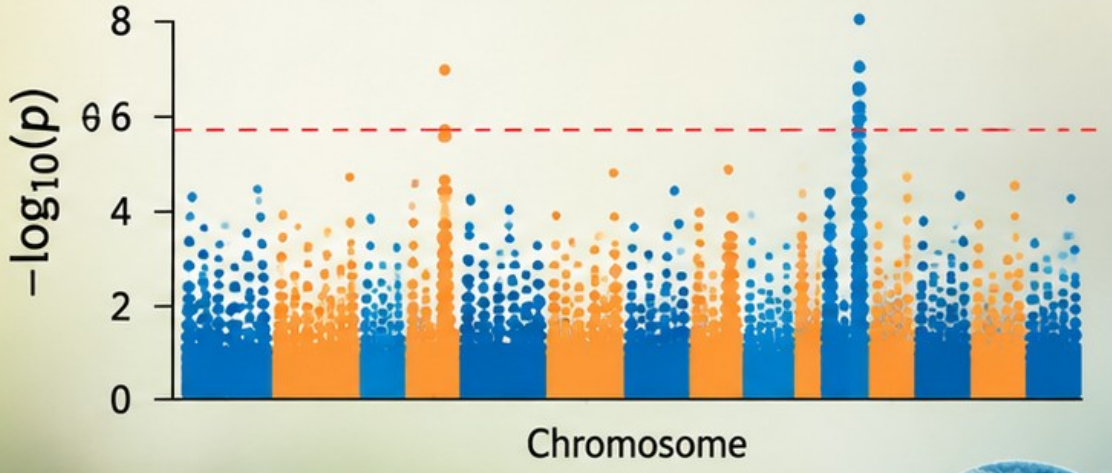
Genome-Wide Association Studies (GWAS) for Evolutionary Biology

Nikolay Oskolkov, Group Leader (PI) at NIRI, Riga, Latvia





Genomics: Progress or Regress?



Genetic variation: the raw material of association

What varies?

Single-nucleotide variants (SNVs / SNPs): one base differs.

Insertions / deletions (indels): bases added or removed.

Structural variation (SVs): larger deletions, duplications, inversions and rearrangements.

Alleles are alternative states; genotype describes the alleles carried by an individual.

How variation is inherited

Nearby variants can be inherited together as a haplotype.

Linkage disequilibrium (LD): non-random association of alleles at different loci.

Recombination breaks down haplotypes over generations.

Allele frequencies and LD patterns differ among populations and species.

Caution of GWAS interpretation

A typed marker may not be causal; it can tag a nearby causal variant in LD

Common variants are well captured by many SNP arrays; rare and structural variants may be missed.

Population history, relatedness and admixture shape allele frequencies.

Reference panels help interpret variation, but must be biologically appropriate.

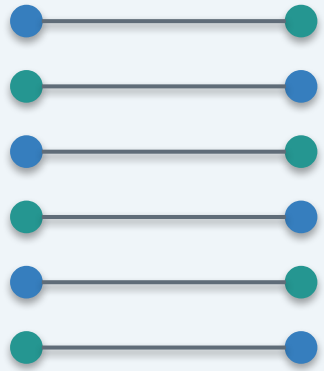
Concepts: NHGRI, International HapMap Project; NHGRI Genomic Variation Program. genome.gov/10001688/international-hapmap-project

CHIP GENOTYPING: test selected positions across the genome

A SNP array does not read every DNA base — it measures predefined markers using allele-specific probes and fluorescence.

1 DNA SAMPLE

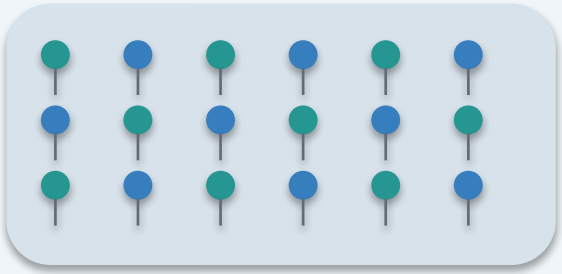
Extract DNA, amplify and fragment it



Many short DNA fragments from across the genome

2 HYBRIDISE TO CHIP

Fragments bind to matching probes at known SNP locations



Each marker has probes that distinguish alternative alleles (e.g. A or G).

3 READ THE SIGNAL

A labelled base is added; a scanner measures fluorescence intensity



AA	AG	GG
homozygous A	heterozygous	homozygous G

Software converts signal ratios into genotype calls: AA, AG or GG

OUTPUT: a genotype matrix — individuals × preselected SNPs (not complete genome sequences).

Principle shown: Illumina Infinium-style SNP array. Other array chemistries differ in detail.

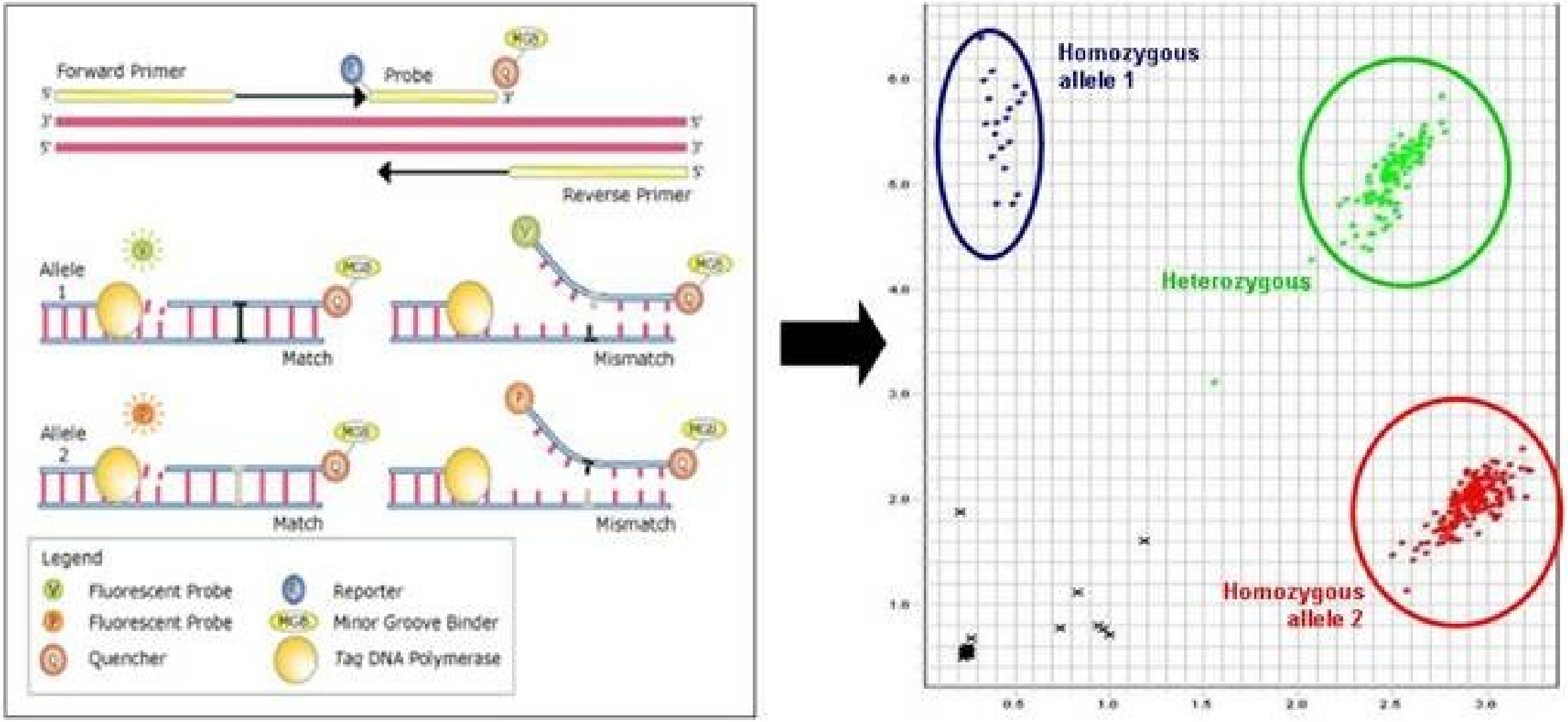
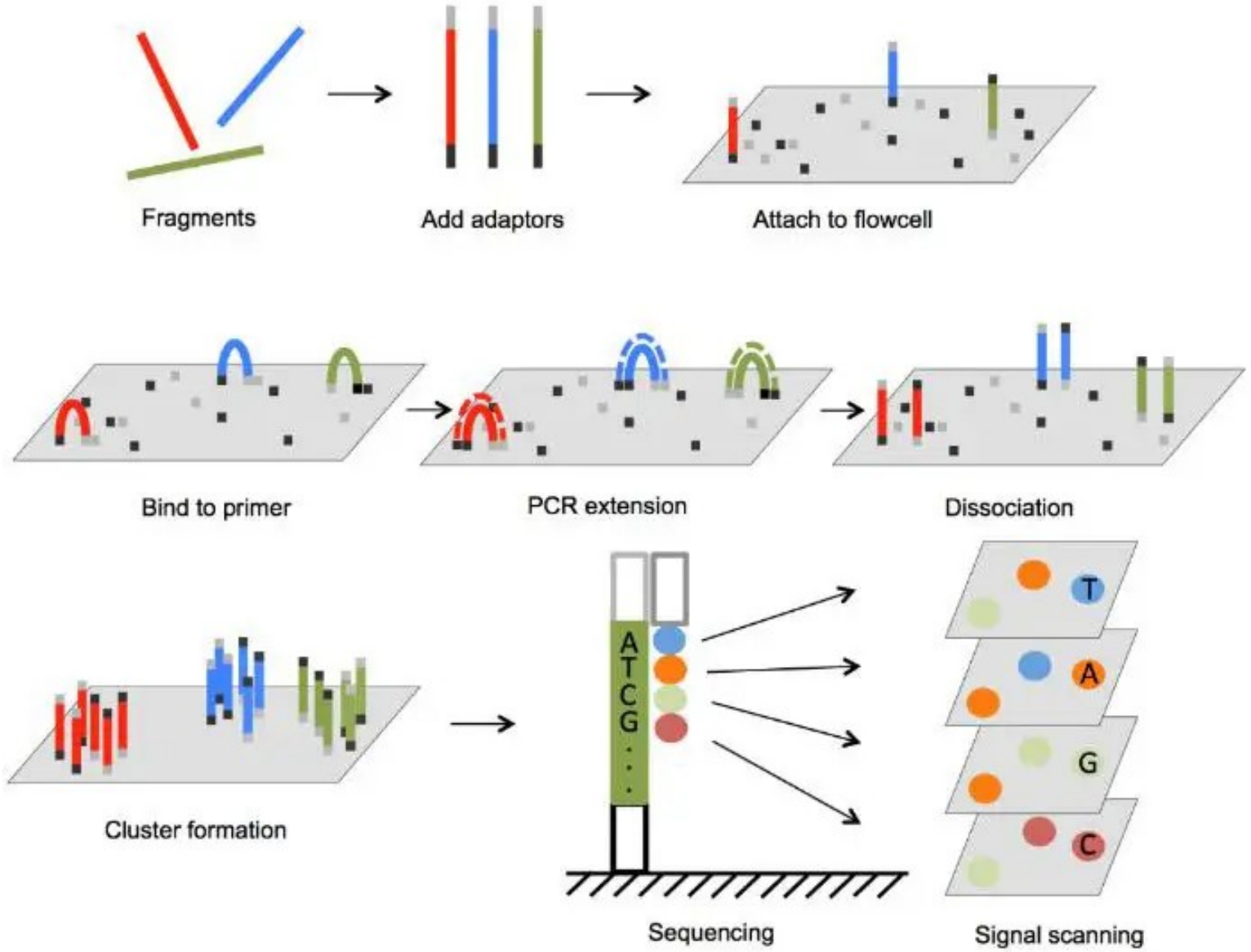
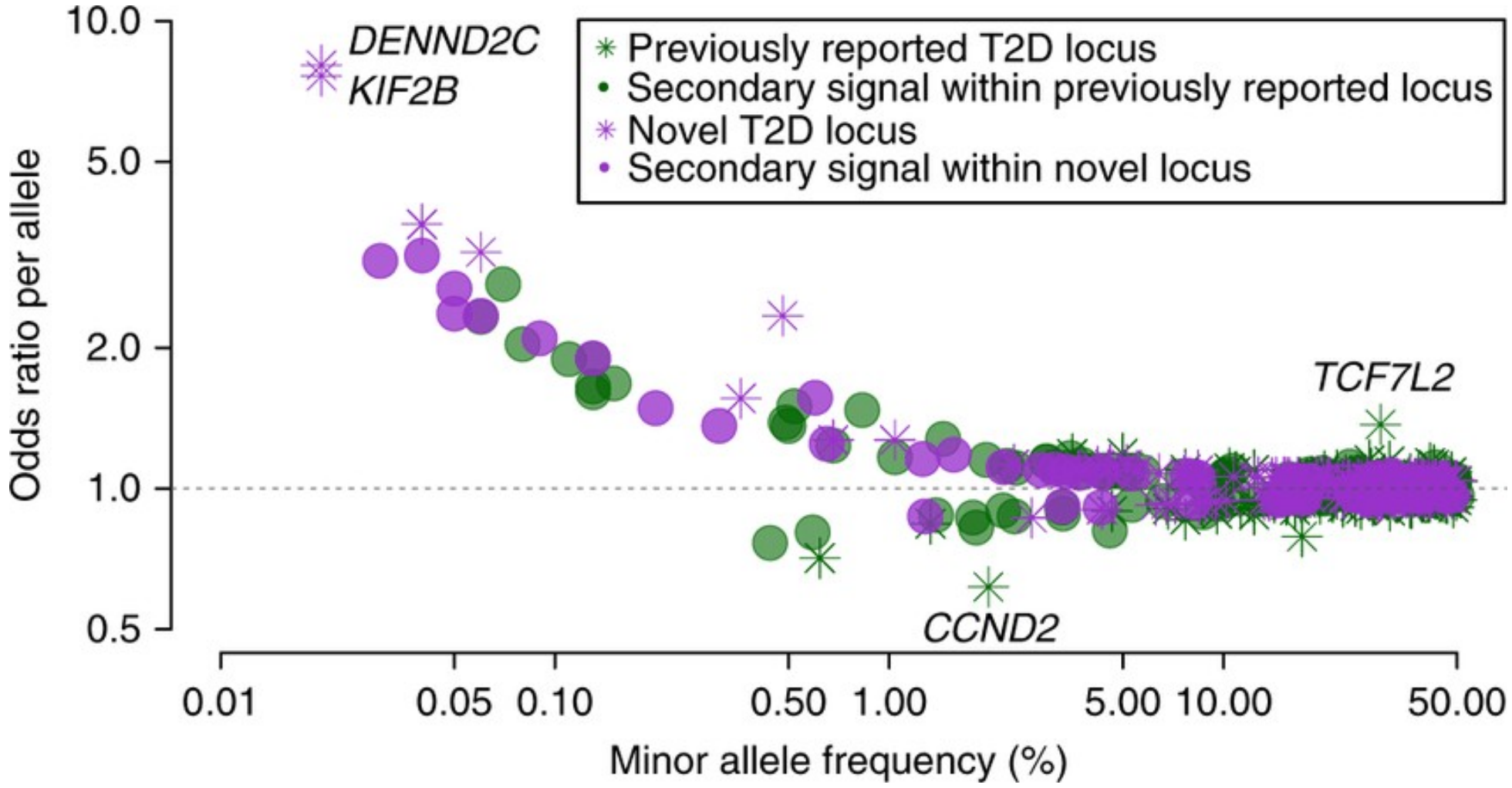


Figure 1: SNP genotyping using Taqman®

Illumina Next Generation Sequencing (NGS)



Rare variants with large effects may be missing



THE BASIC IDEA

GWAS: scan the genome for trait-associated markers



What is tested?

For each variant, test whether genotype dosage predicts the phenotype.

Adjust for relevant covariates, technical effects, ancestry and relatedness.

Millions of tests require stringent multiple-testing control (often $P < 5 \times 10^{-8}$ in human GWAS).

What does a hit mean?

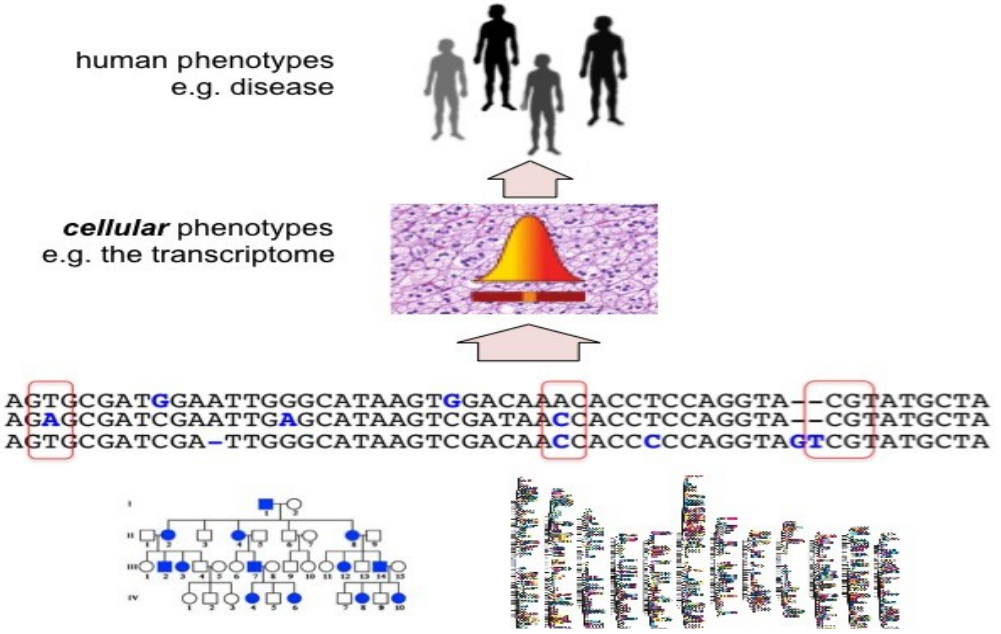
A significant marker identifies an associated region, i.e. not necessarily the causal change.

Follow-up: fine-mapping, functional annotation, replication and experimental validation.

Population structure, batch effects and phenotype error can create misleading signals.

Sources: NHGRI GWAS Fact Sheet; NHGRI-EBI GWAS Catalog training. genome.gov/about-genomics/fact-sheets/Genome-Wide-Association-Studies-Fact-Sheet

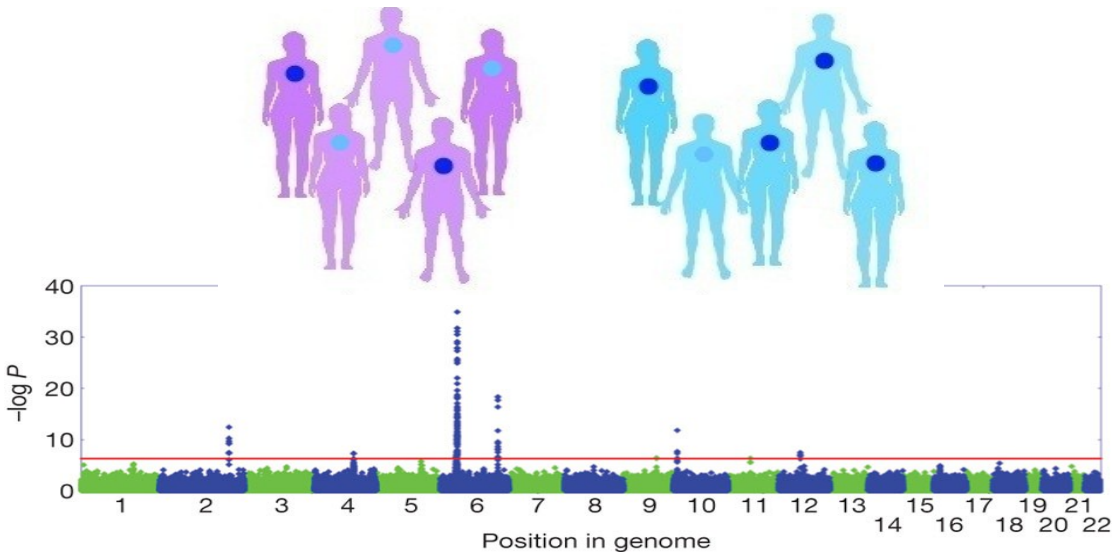
Human Medical Genetics of Diabetes: Search for Genes



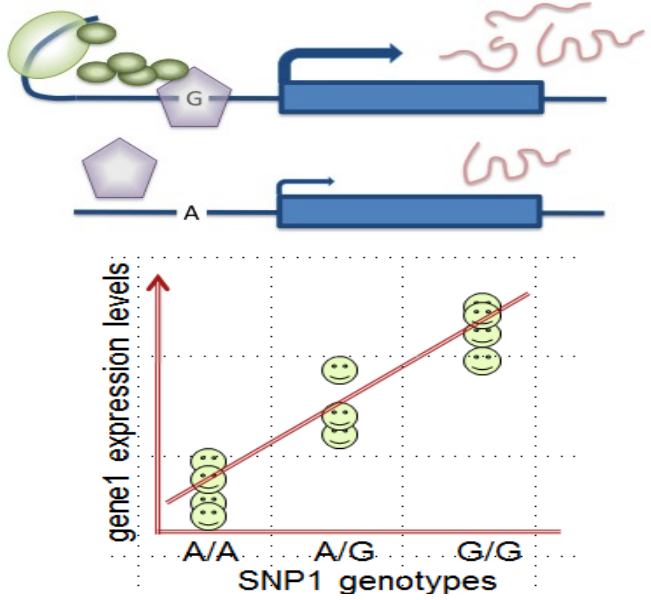
Find a genetic variation in a gene causing a desease via regulating the gene expression

Genome-Wide Association Studies

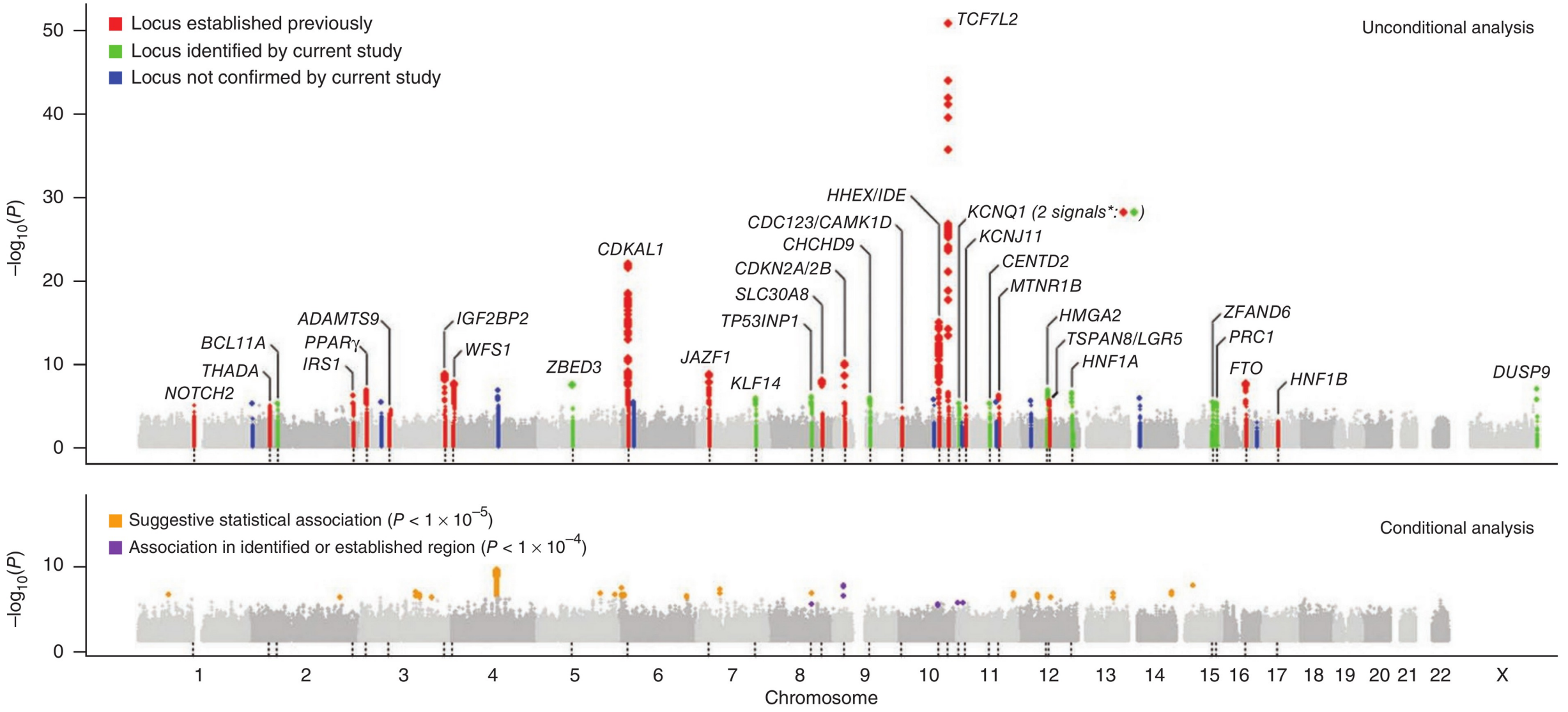
Controls Cases



eQTL Analysis (cis, trans, ASE etc.)



Manhattan plot visualization of GWAS results



SNP Does not Mean Gene

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ORIGINAL ARTICLE

FTO Obesity Variant Circuitry and Adipocyte Browning in Humans

Melina Claussnitzer, Ph.D., Simon N. Dankel, Ph.D., Kyoung-Han Kim, Ph.D., Gerald Quon, Ph.D., Wouter Meuleman, Ph.D., Christine Haugen, M.Sc., Viktoria Glunk, M.Sc., Isabel S. Sousa, M.Sc., Jacqueline L. Beaudry, Ph.D., Vijitha Puviindran, B.Sc., Nezar A. Abdennur, M.Sc., Jannel Liu, B.Sc., Per-Arne Svensson, Ph.D., Yi-Hsiang Hsu, Ph.D., Daniel J. Drucker, M.D., Gunnar Mellgren, M.D., Ph.D., Chi-Chung Hui, Ph.D., Hans Hauner, M.D., and Manolis Kellis, Ph.D.

N Engl J Med 2015; 373:895-907 | September 3, 2015 | DOI: 10.1056/NEJMoa1502214

Comments open through September 9, 2015

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Abstract | Article | References | Citing Articles (140) | Comments (1) | Letters | Metrics

BACKGROUND

Genomewide association studies can be used to identify disease-relevant genomic regions, but interpretation of the data is challenging. The *FTO* region harbors the strongest genetic association with obesity, yet the mechanistic basis of this association remains elusive.

Full Text of Background...

METHODS

We examined epigenomic data, allelic activity, motif conservation, regulator expression, and gene coexpression patterns, with the aim of dissecting the regulatory circuitry and mechanistic basis of the association between the *FTO* region and obesity. We validated our predictions with the use of directed perturbations in samples from patients and from mice and with endogenous CRISPR–Cas9 genome editing in samples from patients.

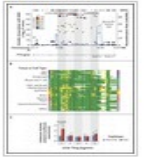
Full Text of Methods...

RESULTS

Our data indicate that the *FTO* allele associated with obesity represses mitochondrial thermogenesis in adipocyte precursor cells in a tissue-autonomous manner. The rs1421085 T-to-C single-nucleotide variant disrupts a conserved motif for the ARID5B repressor, which leads to derepression of a potent preadipocyte enhancer and a doubling of *IRX3* and *IRX5* expression during early adipocyte differentiation. This results in a cell-autonomous developmental shift from energy-dissipating beige (brite) adipocytes to energy-storing white adipocytes, with a reduction in mitochondrial

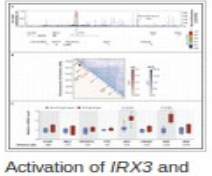
MEDIA IN THIS ARTICLE

FIGURE 1



Activation of a Superenhancer in Human Adipocyte Progenitors by the *FTO* Obesity Risk Haplotype.

FIGURE 2



Activation of *IRX3* and *IRX5* Expression in Human Adipocyte Progenitors by the *FTO* Obesity Risk Genotype.

ARTICLE ACTIVITY

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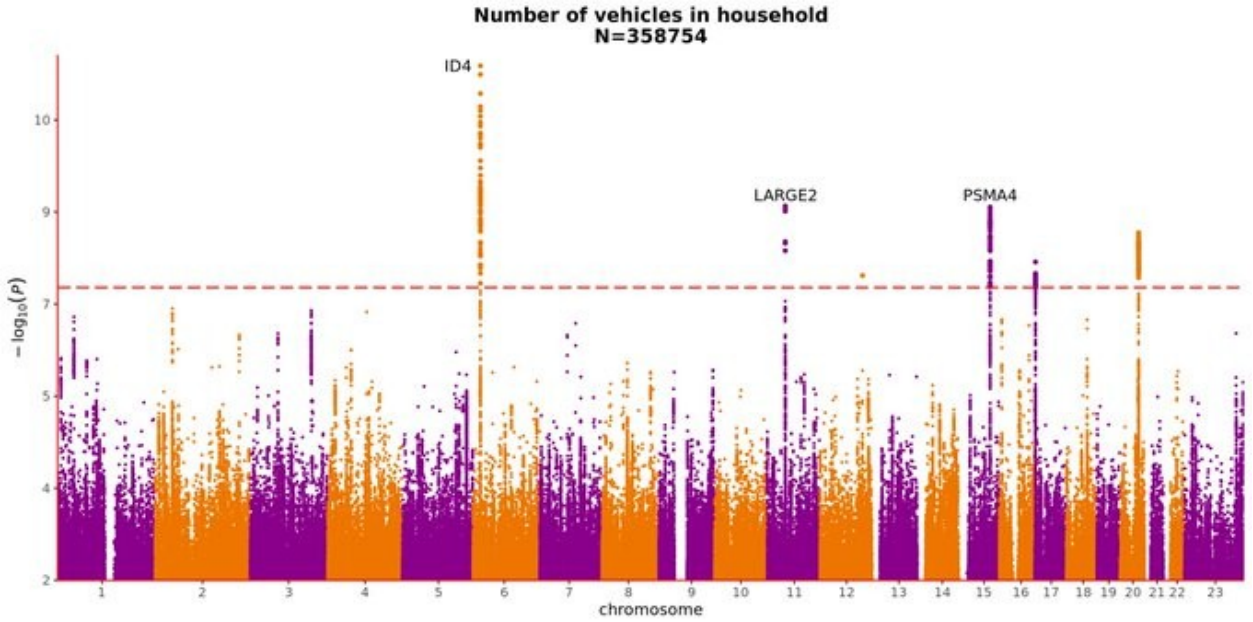
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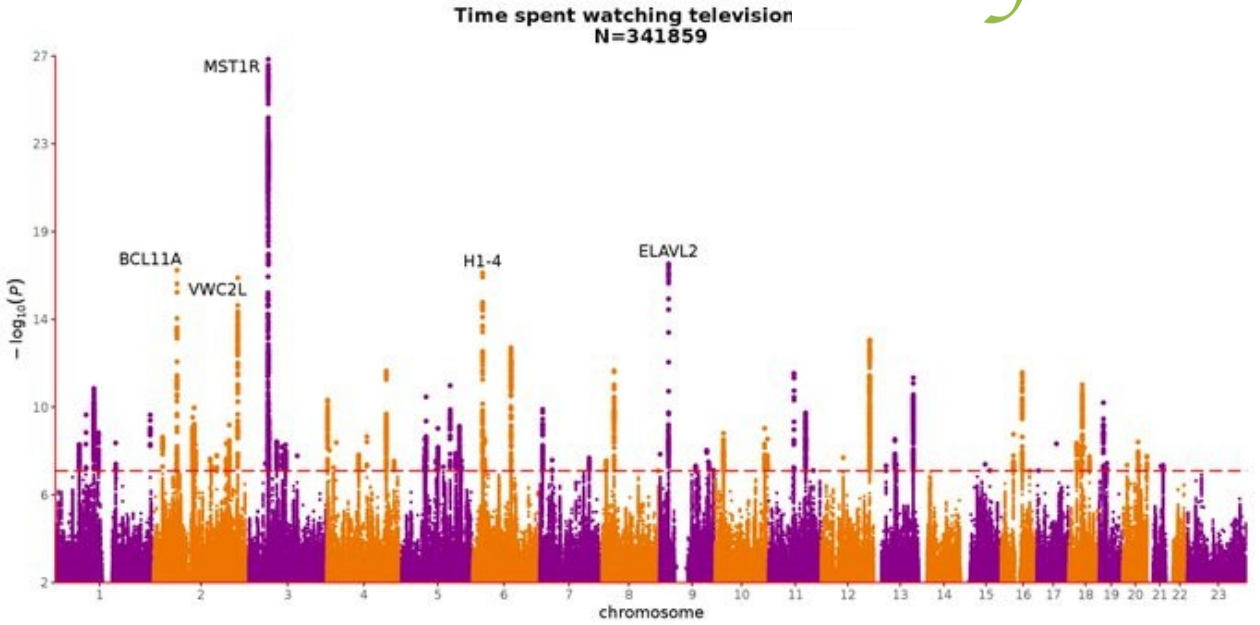
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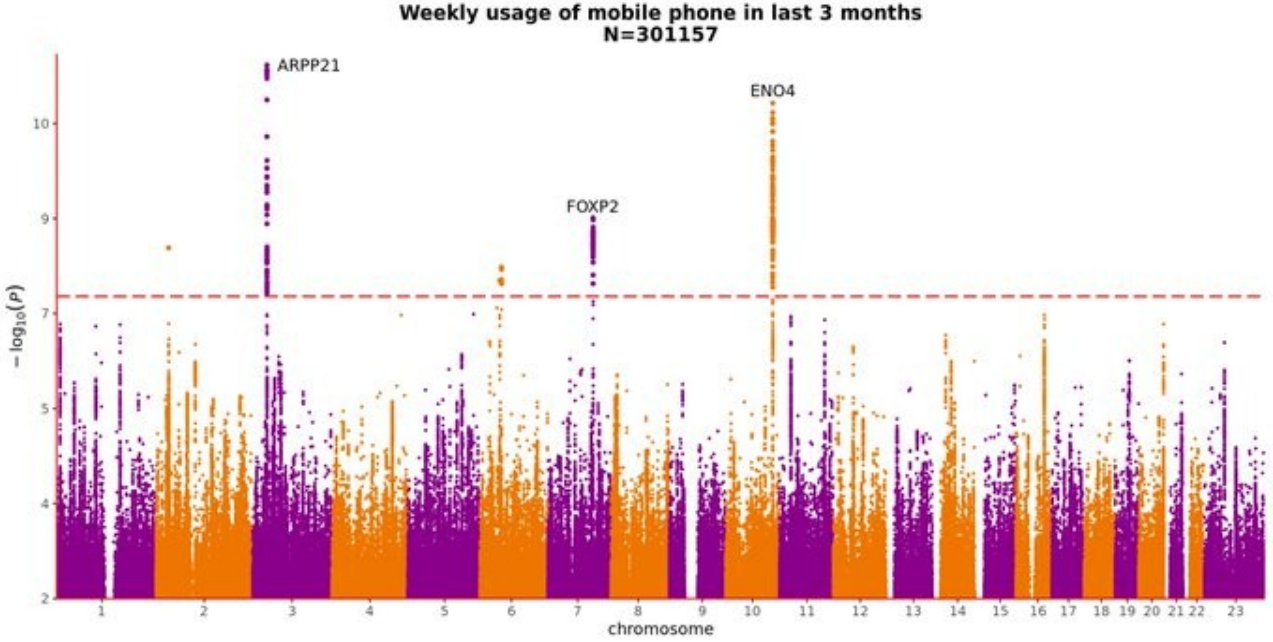
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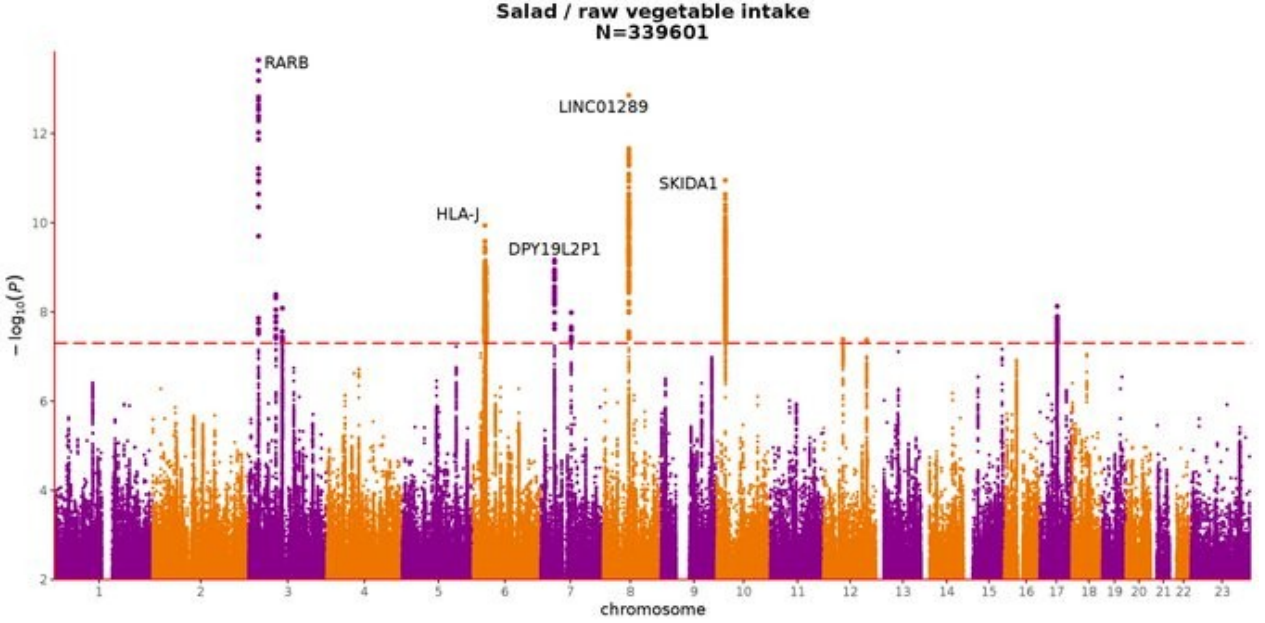
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False-positive hits in GWAS

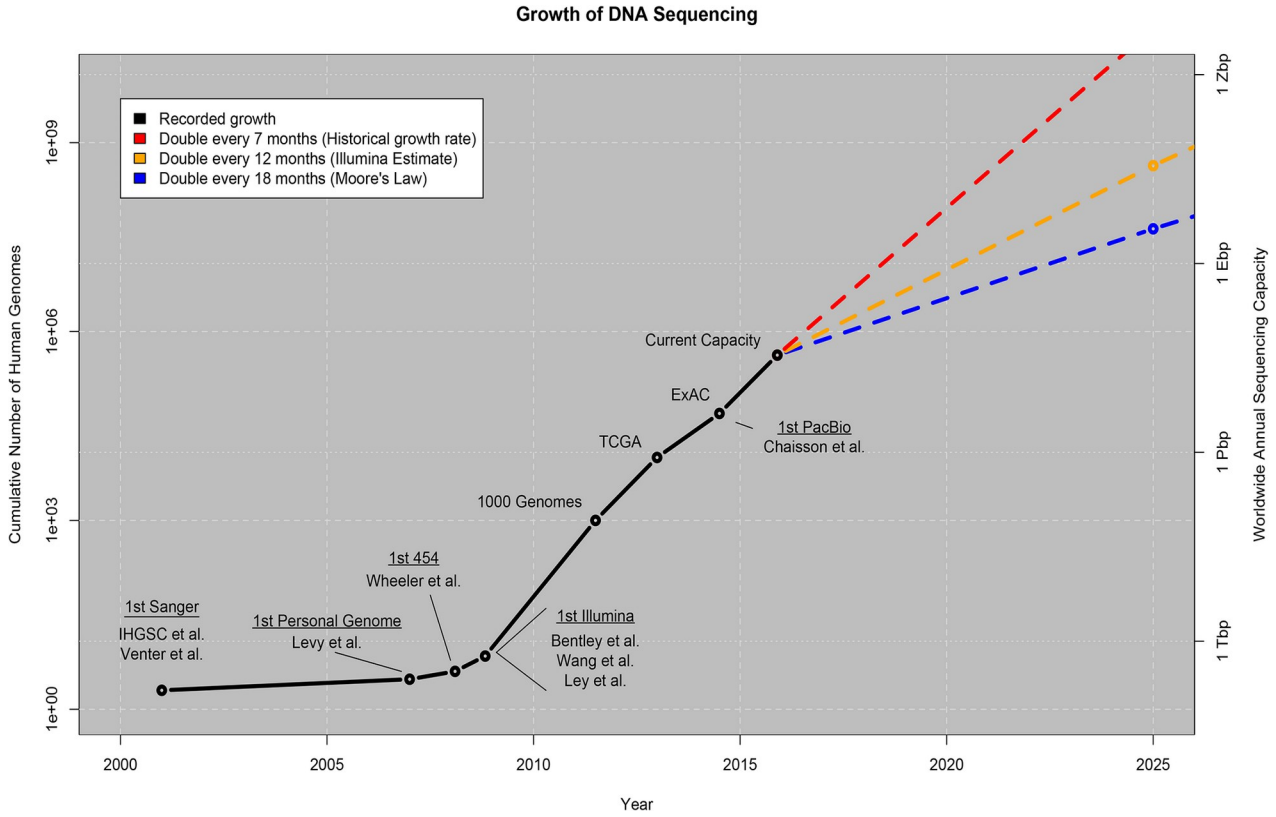




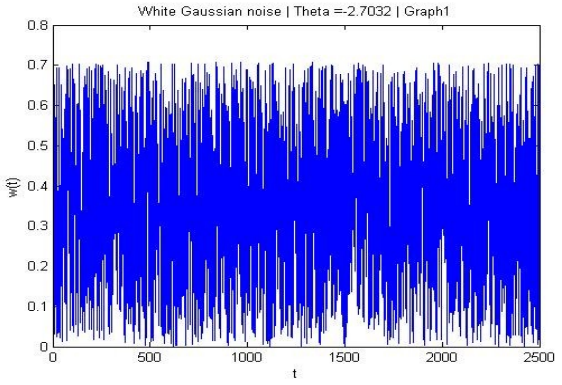




Big Data in Genomics



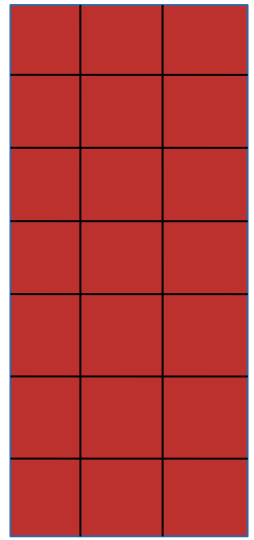
Stephens et al., (2015). Big Data: Astronomical or Genomical? PLoS Biology 13(7)



A file with **White Noise** can also take a lot of disk space

Genomics / WGS : **Little Data**

$N_1 \sim 10^3$

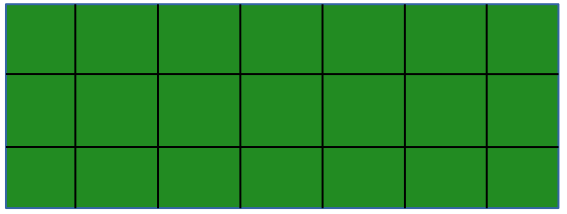


$P_1 \sim 10^6$

$N_1 * P_1 = N_2 * P_2 = 10^9$

scRNAseq: **Big Data**

$N_2 \sim 10^6$



$P_2 \sim 10^3$

International HapMap Project: mapping common haplotypes

What was HapMap?

International effort launched in 2002 to describe common patterns of human DNA variation.

Central product: a genome-wide map of haplotypes and linkage disequilibrium across sampled populations.

Identified 'tag SNPs' that capture much of the information in nearby correlated variants.

Goal: make genome-wide association scans practical with fewer genotyped markers.

Why it mattered

Helped design SNP arrays.

Provided a framework for early GWAS.

Made LD and haplotype structure central to association mapping.

Important caveat

Not a complete catalogue of all human variation.

Sampling was limited compared with global diversity.

Tagging performance depends on ancestry: LD patterns are not universal.

Source: NHGRI, International HapMap Project and project fact sheet. genome.gov/10001688/international-hapmap-project

1000 Genomes and HGDP: complementary views of human diversity

1000 Genomes Project (1KGP)

Public catalogue of human variation built using whole-genome sequencing, exome sequencing and genotyping.

Final Phase 3: 2,504 people from 26 populations; >88 million variants, with phased haplotypes.

Designed to capture common variants broadly; useful for allele frequencies, phasing and genotype imputation.

A widely used human reference panel—but not universal for every ancestry or species.

Human Genome Diversity Project (HGDP)

Collection created to study the breadth and history of human genetic diversity across populations worldwide.

Classic HGDP-CEPH panel: about 1,050 individuals from 52 populations (counts vary slightly by dataset/marker set).

Historically important for population structure, migration, admixture and human evolutionary history.

Compared with 1KGP, HGDP emphasizes broad geographic / population diversity rather than a large, uniform imputation panel.

Sources: 1000 Genomes Project Consortium (2015), Nature; IGSR summary; Stanford Rosenberg Lab HGDP resources. internationalgenome.org/1000-genomes-summary/

European Reference Genome Atlas (ERGA): reference genomes for biodiversity

What is ERGA?

Pan-European scientific community coordinating high-quality reference genomes for European biodiversity.

European node of the Earth BioGenome Project, which aims to build reference genomes across Earth's eukaryotic biodiversity.

Pilot tested a distributed model: 98 species across 33 European countries.

Work spans sampling, sequencing, assembly, annotation, data stewardship, training and shared standards.

Relevance for evolutionary biology genomics

A reference genome is the coordinate system for read mapping, variant calling and annotation.

High-quality assemblies improve detection of SNPs, indels and structural variation.

Comparative and population genomics can connect variation to adaptation, demography and conservation.

ERGA is not a human diversity panel: it supports species-specific genomics across biodiversity.

Sources: ERGA official FAQ; ERGA pilot paper (npj Biodiversity, 2024). erga-biodiversity.eu/faqs

TAKE-HOME COMPARISON

Four resources, different questions and different scales

Resource	Scope	Core contribution	Useful for
HapMap	Human; common variation	Haplotype / LD map; tag SNPs	Array design; early GWAS; understanding LD
1000 Genomes	Human; 26 populations	Broad variant catalogue + phased haplotypes	Allele frequencies; phasing; imputation
HGDP	Human; 52 populations	Population-diversity panel with broad geographic sampling	Population structure; ancestry; human evolution
ERGA	European biodiversity; many species	Collaborative production of high-quality species reference genomes	Read mapping; variant discovery; evolution; conservation

Key principle: choose a reference resource that matches the biological question, species and population / taxonomic diversity.

Links: HapMap — genome.gov/10001688/international-hapmap-project | 1KGP — internationalgenome.org | HGDP — rosenberglab.stanford.edu/diversity.html | ERGA — erga-biodiversity.eu